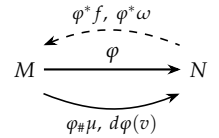


Push-forward and pullback

How integrals travel along a map $\varphi: M \rightarrow N$: functions and forms pull back, measures and vectors push forward, and the Jacobian is the exchange rate. Change of variables, the area formula, push-forward measures and their densities, the coarea formula, and the conventions that bite.



solid: pushed forward dashed: pulled back

1 WHICH WAY THINGS MOVE

A map $\varphi: M \rightarrow N$ sends points forward. Anything that is *evaluated* at points of N (a function, a covector, a k -form) can be composed with φ and so comes back to M : it pulls back. Anything that is *made of* points of M (a curve, a tangent vector, a mass distribution) can be carried along: it pushes forward. Integration pairs the two, and the pairing is preserved.

pull back	$\varphi^* f = f \circ \varphi,$ $(\varphi^* \omega)_p(v_1, \dots, v_k) = \omega_{\varphi(p)}(d\varphi v_1, \dots, d\varphi v_k)$
push forward	$d\varphi_p(v) = \frac{d}{dt} \varphi(\gamma(t)) _0, \quad (\varphi_\# \mu)(B) = \mu(\varphi^{-1} B)$
duality	$\int_N g d(\varphi_\# \mu) = \int_M (\varphi^* g) d\mu, \quad \int_{\varphi \circ \gamma} \omega = \int_\gamma \varphi^* \omega$
functoriality	$(\psi \circ \varphi)^* = \varphi^* \psi^* \text{ (order reverses),}$ $(\psi \circ \varphi)_\# = \psi_\# \varphi_\#$

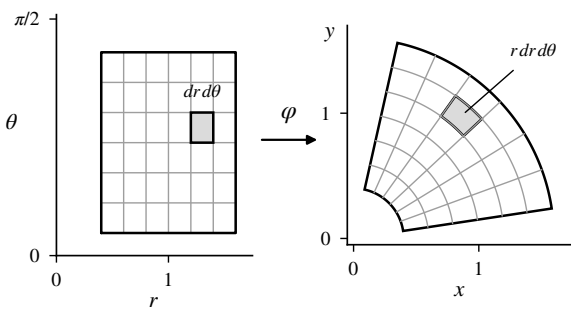
Pullback works for *any* smooth map. Push-forward of a vector field, or of a function, needs φ to be a diffeomorphism; a measure always pushes forward, and a function can be pushed forward only by integrating over the fibres (§6).

2 PULLING BACK FORMS

In coordinates $y = \varphi(x)$, replace every y^i by $\varphi^i(x)$ and every dy^i by its differential:

$$\begin{aligned} \varphi^*(dy^i) &= d\varphi^i = \sum_j \frac{\partial \varphi^i}{\partial x^j} dx^j \\ \varphi^*(\alpha \wedge \beta) &= \varphi^* \alpha \wedge \varphi^* \beta, \quad d\varphi^* = \varphi^* d \\ \varphi^*(f dy^1 \wedge \dots \wedge dy^n) &= (f \circ \varphi) \det D\varphi dx^1 \wedge \dots \wedge dx^n \end{aligned}$$

The top-degree rule is the whole reason Jacobians appear: $dy^1 \wedge \dots \wedge dy^n$ is alternating and multilinear, so the $n \times n$ matrix $D\varphi$ acts on it by its determinant. For k -forms, the coefficient is a $k \times k$ minor: $\varphi^*(dy^I) = \sum_J \det(\partial \varphi^I / \partial x^J) dx^J$.



Polar coordinates $\varphi(r, \theta) = (r \cos \theta, r \sin \theta)$. A cell $dr d\theta$ lands on a curved cell of area $r dr d\theta$, because $\varphi^*(dx \wedge dy) = (\cos \theta dr - r \sin \theta d\theta) \wedge (\sin \theta dr + r \cos \theta d\theta) = r dr \wedge d\theta$.

Jacobians to know

map	$ \det D\varphi $ or area element
polar (r, θ)	r
cylindrical (r, θ, z)	r
spherical (r, θ, ϕ) , θ from the pole	$r^2 \sin \theta$
linear $x \mapsto Ax + b$	$ \det A $
n -ball in polar form	$r^{n-1} dr d\sigma_{n-1}$
graph $z = h(x, y)$	$\sqrt{1 + \nabla h ^2}$
surface $\varphi(u, v)$ in $\mathbb{R}^3$	$ \varphi_u \times \varphi_v $
k -surface $\varphi: U \subset \mathbb{R}^k \rightarrow \mathbb{R}^n$	$\sqrt{\det(D\varphi^T D\varphi)}$

3 INTEGRATING FORMS: THE SIGN IS KEPT

For an orientation-preserving diffeomorphism $\varphi: M \rightarrow N$ of oriented n -manifolds and a compactly supported n -form ω on N ,

$$\int_M \varphi^* \omega = \int_N \omega, \quad \text{in general} \quad \int_M \varphi^* \omega = \deg \varphi \int_N \omega$$

for a proper smooth map between connected oriented manifolds. The degree counts preimages of a regular value with signs, $\deg \varphi = \sum_{x \in \varphi^{-1}(y)} \text{sgn} \det D\varphi(x)$. Reversing orientation flips the sign. Line integrals are the case $n = 1$: $\int_\gamma \omega = \int_a^b \omega_{\gamma(t)}(\gamma'(t)) dt$, unchanged by any orientation-preserving reparametrisation.

Winding number On $\mathbb{R}^2 \setminus 0$, $d\theta = (x dy - y dx) / (x^2 + y^2)$ is closed but not exact. For a loop $\gamma: S^1 \rightarrow \mathbb{R}^2 \setminus 0$, $\frac{1}{2\pi} \int_{S^1} \gamma^* d\theta = \deg(\gamma / |\gamma|)$, an integer, because $\gamma^* d\theta$ is the pullback of the angle form on S^1 , whose total is 2π .

Stokes commutes with pullback: $\int_M d\varphi^* \eta = \int_{\partial M} \varphi^* \eta$, so the signed integral of $\det D\varphi$ over a region depends only on where the boundary goes. For a map of the square with straight image edges, it is the shoelace area of the four corners, fold or no fold.

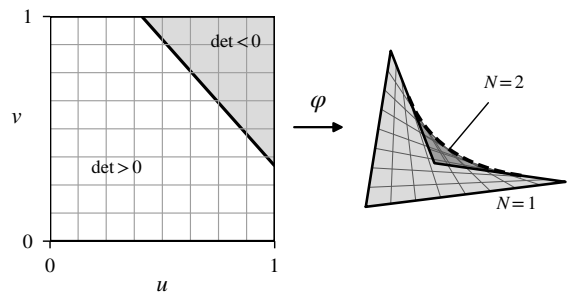
4 MEASURES: THE ABSOLUTE VALUE

Lebesgue measure does not see orientation, so the determinant enters with its absolute value. If $\varphi: U \rightarrow \varphi(U)$ is a C^1 diffeomorphism of open sets in $\mathbb{R}^n$ and $f \geq 0$ is measurable,

$$\int_{\varphi(U)} f(y) dy = \int_U f(\varphi(x)) |\det D\varphi(x)| dx.$$

The area formula Drop injectivity. For φ Lipschitz (in particular C^1) on $U \subset \mathbb{R}^n$ into $\mathbb{R}^n$, each point is counted as often as it is hit:

$$\begin{aligned} \int_U f(\varphi(x)) |\det D\varphi(x)| dx &= \int_{\mathbb{R}^n} f(y) N(\varphi, y) dy \\ N(\varphi, y) &= \#(\varphi^{-1}(y) \cap U) \end{aligned}$$



A bilinear map of the square with one corner pulled inside the other three. $\det D\varphi$ is affine in (u, v) , so it changes sign across a straight line (left). Its image (dashed) is the fold, and the sliver beyond it is covered twice. With $|\det|$ the pulled-back integral counts that sliver twice; with the signed $\det$ it cancels it.

When M is k -dimensional inside $\mathbb{R}^n$ ($k < n$), the determinant is replaced by the Gram factor $J_k \varphi = \sqrt{\det(D\varphi^T D\varphi)}$, the k -volume of the parallelepiped spanned by the partial derivatives; the area formula then computes Hausdorff measure $\mathcal{H}^k$ of the image, with multiplicity.

Gaussian integral

$(\int_{\mathbb{R}} e^{-x^2} dx)^2 = \int_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy = \int_0^{2\pi} \int_0^\infty e^{-r^2} r dr d\theta = \pi$. The polar map fails to be injective only on a null set (the ray $\theta = 0$ and the origin), which the formula ignores.

5 PUSH-FORWARD OF A MEASURE

For a measurable $T: X \rightarrow Y$ and a measure μ on X , the image measure is $T_\# \mu(B) = \mu(T^{-1}B)$, also written $T_* \mu$ or $\mu \circ T^{-1}$. In probability: if $X \sim \mu$ then $T(X) \sim T_\# \mu$. Every property is inherited from the one line

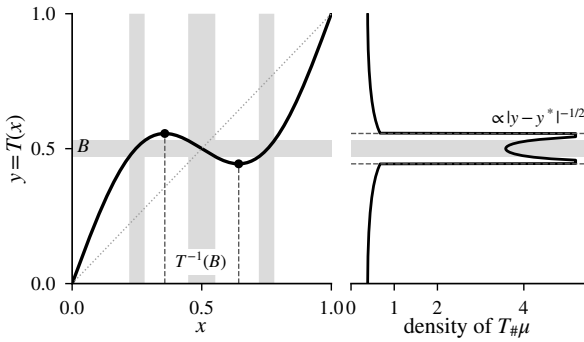
$$\int_Y g d(T_\# \mu) = \int_X (g \circ T) d\mu$$

for all measurable $g \geq 0$, or $g \in L^1(T_\# \mu)$

Proof: it holds for indicators by definition, hence for simple functions, hence by monotone convergence for all $g \geq 0$. Consequences: total mass is preserved; $T_\# \delta_x = \delta_{T(x)}$; empirical measures push to empirical measures, $T_\# (\frac{1}{n} \sum \delta_{x_i}) = \frac{1}{n} \sum \delta_{T(x_i)}$; expectations are computed upstairs, $\mathbb{E}[g(T(X))] = \int g \circ T d\mu$.

The density formula If $\mu = \rho dx$ on $\mathbb{R}^n$ and T is C^1 with $\det DT \neq 0$ almost everywhere, then $T_\# \mu$ has density

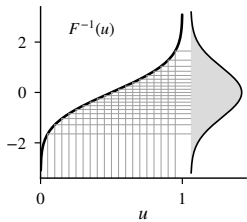
$$\begin{aligned} \rho_{T_\# \mu}(y) &= \sum_{x \in T^{-1}(y)} \frac{\rho(x)}{|\det DT(x)|} \\ &= \rho(T^{-1}y) |\det D(T^{-1})(y)| \quad \text{if } T \text{ is injective} \end{aligned}$$



$T(x) = x + a \sin(2\pi x)/2\pi$ with $a = 1.6$ and μ uniform on $[0, 1]$. The window B pulls back to three pieces. At the fold heights y^* , where $T' = 0$, the density has integrable $|y - y^*|^{-1/2}$ singularities; the measure itself stays perfectly finite.

One-dimensional recipes

$Y = T(X)$	density of Y
$aX + b$	$\rho((y-b)/a)/ a $
X^2	$(\rho(\sqrt{y}) + \rho(-\sqrt{y})) / (2\sqrt{y}), y > 0$; for $X \sim N(0, 1)$ this is χ_1^2
e^X	$\rho(\log y)/y$ (log-normal from normal)
$F_X(X)$	uniform on $[0, 1]$ if F_X is continuous
$F^{-1}(U), U \sim \text{Unif}$	law with CDF F (inverse-CDF sampling)
$\tan(\pi(U - \frac{1}{2}))$	Cauchy: $1/\pi(1+y^2)$



The quantile function F^{-1} pushes Lebesgue measure on $[0, 1]$ to μ : equal steps in u become steps in y that crowd where the density is high. This is the push-forward read backwards: $(F^{-1})_\# \text{Leb}_{[0,1]} = \mu$.

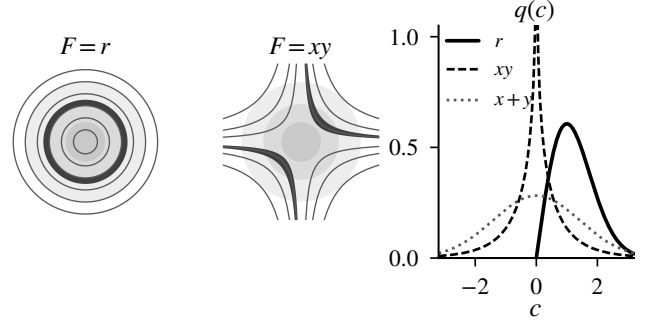
Box-Muller Push the uniform measure on $(0, 1)^2$ forward by $(u_1, u_2) \mapsto \sqrt{-2 \log u_1} (\cos 2\pi u_2, \sin 2\pi u_2)$. In polar form

$r = \sqrt{-2 \log u_1}, \theta = 2\pi u_2$, so $u_1 = e^{-r^2/2}$ and $|du_1 du_2| = \frac{1}{2\pi} r e^{-r^2/2} dr d\theta = \frac{1}{2\pi} e^{-(x^2+y^2)/2} dx dy$: two independent standard normals.

6 INTEGRATING ALONG THE FIBRES

When $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ with $m < n$, the preimage of a point is a fibre of dimension $n - m$, and the push-forward density is an integral over it. The weight is the reciprocal of the normal Jacobian $JF = \sqrt{\det(DF DF^T)}$ (for $m = 1$, simply $|\nabla F|$), because the shell between the fibres at c and $c + \delta$ has thickness $\delta/|\nabla F|$.

$$\begin{aligned} \text{coarea: } \int_{\mathbb{R}^n} g JF dx &= \int_{\mathbb{R}^m} \left(\int_{F^{-1}(c)} g d\mathcal{H}^{n-m} \right) dc \\ \text{density of } F(X): \quad q(c) &= \int_{F^{-1}(c)} \frac{p}{JF} d\mathcal{H}^{n-m} = \frac{dF_\#(p dx)}{dc} \end{aligned}$$



A standard Gaussian on the plane with its fibres, and one shell $c \leq F \leq c + \delta$ in dark grey. For $F = xy$ the shell near $c = 0$ hugs the axes, where $|\nabla F|$ is small, so it is wide and q has a logarithmic spike, $K_0(|c|)/\pi$.

F	$ \nabla F $	$q(c)$
x	1	marginal $\int p(c, y) dy$
$x + y$	$\sqrt{2}$	convolution $\int p(t, c - t) dt$ (the $\sqrt{2}$ from $d\ell$ cancels)
$\sqrt{x^2 + y^2}$	1	$c \int_0^{2\pi} p(c \cos \theta, c \sin \theta) d\theta$; Gaussian: Rayleigh $c e^{-c^2/2}$
xy	$\sqrt{x^2 + y^2}$	$\int p(x, c/x) \frac{dx}{ x }$; Gaussian: $K_0(c)/\pi$
y/x	$\sqrt{x^2 + y^2}/x^2$	$\int p(x, cx) x dx$; Gaussian: Cauchy

Fibre integration of forms For a bundle $\pi: E \rightarrow B$ with compact oriented fibres of dimension m , integration over the fibre is a push-forward $\pi_*: \Omega^k(E) \rightarrow \Omega^{k-m}(B)$ with $\int_B \pi_* \omega = \int_E \omega$, $d\pi_* = \pi_* d$ (closed fibres), and the projection formula $\pi_*(\pi^* \alpha \wedge \beta) = \alpha \wedge \pi_* \beta$. Measures and top forms push; lower forms push only with fibres to integrate over.

7 FLOWS: PUSH-FORWARD IN TIME

Let Φ_t be the flow of a vector field X and $\mu_t = (\Phi_t)_\# \mu_0$ with density ρ_t . Differentiating $\int g d\mu_t = \int g \circ \Phi_t d\mu_0$ gives the continuity equation, and pulling back gives the Lie derivative:

$$\begin{aligned} \partial_t \rho + \nabla \cdot (\rho X) &= 0, & \frac{d}{dt} \det D\Phi_t &= (\nabla \cdot X) \circ \Phi_t \det D\Phi_t \\ \frac{d}{dt} \Big|_0 \Phi_t^* \omega &= \mathcal{L}_X \omega = d \iota_X \omega + \iota_X d\omega \end{aligned}$$

Liouville: a divergence-free field ($\mathcal{L}_X \text{vol} = 0$) preserves volume, so densities are simply transported, $\rho_t = \rho_0 \circ \Phi_t^{-1}$. Hamiltonian flows preserve ω^n in phase space for this reason.

8 CONVENTIONS AND TRAPS

- **Sign or absolute value.** Forms use $\det D\varphi$ and need orientations; measures and densities use $|\det D\varphi|$ and do not. On a non-orientable manifold only densities integrate.
- **Which Jacobian.** In $\int f(y) dy = \int f(\varphi(x)) |\det D\varphi(x)| dx$ the Jacobian is of the map *from* the integration variable. The density of $T(X)$ divides by $|\det DT|$ at the preimage, equivalently multiplies by $|\det D(T^{-1})|$ at y .
- **All the preimages.** A non-injective map gives a sum over $T^{-1}(y)$; forgetting a branch (as with X^2) loses mass.
- **Order reverses.** $(\psi \circ \varphi)^* = \varphi^* \psi^*$: pullback is contravariant. The matrix of φ^* on covectors is the transpose $D\varphi^T$.
- **Notation.** φ_* means the differential in geometry, the image measure in probability, and fibre integration for forms; $T_{\#}\mu$ is unambiguous. Some texts write $\varphi^*\mu$ for the pullback of a measure along a local diffeomorphism, which composes the density with φ and multiplies by $|\det D\varphi|$.
- **Critical values.** Where $\nabla F = 0$ on a fibre, q may blow up (the xy spike, the $|y - y^*|^{-1/2}$ folds) while its integral stays finite. By Sard, critical values are a null set.

9 WORKED EXAMPLES

Recipe for the law of $T(X)$

1. If T is not square ($\mathbb{R}^n \rightarrow \mathbb{R}^m$, $m < n$), complete it to a square map with auxiliary coordinates, or go straight to the coarea formula (§6).
2. Solve $T(x) = y$: list every branch of the inverse.
3. On each branch multiply ρ by $|\det D(T^{-1})|$, sum, then integrate out the auxiliary coordinates.
4. Check: total mass 1, and one moment against $\int g \circ T d\mu$.

Area of the sphere $\varphi(\theta, \phi) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ has $|\varphi_\theta \times \varphi_\phi| = \sin \theta$, so the area is $\int_0^{2\pi} \int_0^\pi \sin \theta d\theta d\phi = 4\pi$. Archimedes' hat-box: the projection to the cylinder $(\phi, z = \cos \theta)$ has $dz \wedge d\phi = -\sin \theta d\theta \wedge d\phi$, so it preserves area up to orientation.

Volume of the unit n -ball Push Lebesgue measure forward by $R(x) = |x|$: $R_{\#}(dx) = |S^{n-1}| r^{n-1} dr$. Then $\pi^{n/2} = \int e^{-|x|^2} dx = |S^{n-1}| \int_0^\infty r^{n-1} e^{-r^2} dr = |S^{n-1}| \Gamma(\frac{n}{2})/2$, so $|S^{n-1}| = 2\pi^{n/2}/\Gamma(\frac{n}{2})$ and $|B^n| = |S^{n-1}|/n = \pi^{n/2}/\Gamma(\frac{n}{2} + 1)$.

Gaussians under linear maps If $X \sim N(m, \Sigma)$ and A is invertible, $(A \cdot + b)_{\#} N(m, \Sigma) = N(Am + b, A\Sigma A^T)$: the density picks up $1/|\det A|$, exactly the factor that turns $\sqrt{\det \Sigma}$ into $\sqrt{\det A\Sigma A^T}$. For a non-square A the same answer follows from the characteristic function, $\mathbb{E} e^{i\tilde{\xi} \cdot AX} = \mathbb{E} e^{i(A^T \tilde{\xi}) \cdot X}$, a pullback of the test function.

Gamma and Beta $X \sim \Gamma(a)$, $Y \sim \Gamma(b)$ independent. Map $(x, y) \mapsto (s, t) = (x + y, x/(x + y))$ with inverse $x = st$, $y = s(1 - t)$ and $|\det| = s$. The joint density becomes $\frac{s^{a+b-1} e^{-s}}{\Gamma(a+b)} \cdot \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} t^{a-1} (1-t)^{b-1}$: $S \sim \Gamma(a+b)$ and $T \sim \text{Beta}(a, b)$ are independent, and $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ drops out.

10 TRANSPORT: CHOOSING THE MAP

Given μ and ν , a transport map is any T with $T_{\#}\mu = \nu$. In one dimension the monotone one is $T = F_\nu^{-1} \circ F_\mu$. In $\mathbb{R}^n$, Brenier: for quadratic cost the optimal map is $T = \nabla \phi$ with ϕ convex, and the density formula of §5 becomes the Monge–Ampère equation

$$\rho_\mu(x) = \rho_\nu(\nabla \phi(x)) \det D^2 \phi(x)$$

$$W_2^2(\mu, \nu) = \int |x - \nabla \phi(x)|^2 d\mu(x)$$

No absolute value is needed: $D^2 \phi$ is positive semidefinite, so the determinant is already ≥ 0 . Normalising flows in machine learning are the same identity, $\log \rho_\mu = \log \rho_\nu \circ T + \log |\det DT|$, with T built so that $\det DT$ is cheap.

SOURCES

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Interactive companion: theisegoria.github.io/pushforward-pullback/